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Library Management System using Python

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1. Project Title

Library Management System using Python

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2. Problem Statement

To design and implement a Library Management System that allows users to manage books, register users, search books by category, borrow books, return books, and maintain borrowing records using a menu-driven approach.

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3. Objective

- To simulate basic library operations using Python
- To register users and store their details
- To organize books in different library rows
- To allow users to borrow and return books
- To maintain borrowing records with dates
- To implement menu-driven programming using loops and conditions

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4. Tools Used

- Python 3
- VS Code / IDLE

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5. Description

This project simulates a simple Library Management System. The system allows the librarian to register users, create rows for arranging books, and add books to the library.

Users can view the list of books, search books by category, borrow books, and return them. When a book is borrowed, the system records the borrow date, and when it is returned, the return date is stored.

The entire system works using a **menu-driven interface**, allowing the user to select different operations such as adding books, borrowing books, or viewing records. Dictionaries are used to store user information, book details, library rows, and borrow records.

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6. Algorithm

1. Start the program
2. Initialize dictionaries for library rows, books, users, and borrow records
3. Display the main menu
4. Register a new user and store user details
5. Setup library rows and store number of books in each row
6. Add books with details such as title, author, category, and row number
7. Display all books available in the library
8. Search books based on category
9. Borrow a book if it is available and store the borrow date
10. Return a book and update the return date
11. View borrow records
12. Repeat the process until the user selects exit

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7. Program (Python Code)

```
# ==============================
#      MINI LIBRARY SYSTEM
# ==============================

import datetime

library_rows = {}
books = {}
users = {}
```

```
borrow_records = {}

# --------------------------------
# 1. Register User
# --------------------------------
def register_user():
    print("\n----- USER REGISTRATION -----")

    user_id = input("Enter User ID: ")
    name = input("Enter User Name: ")
    phone = input("Enter Phone Number: ")

    users[user_id] = {
        "name": name,
        "phone": phone
    }

    print("User registered successfully!\n")


# --------------------------------
# 2. Setup Library Rows
# --------------------------------
def setup_library_rows():
    print("\n----- LIBRARY ROW SETUP -----")

    rows = int(input("Enter number of rows in library: "))

    for i in range(1, rows + 1):
        count = int(input(f"How many books in Row {i}: "))
        library_rows[i] = count

    print("Library rows created successfully!\n")


# --------------------------------
# 3. Add Books
# --------------------------------
def add_books():
    print("\n----- ADD BOOKS -----")
```

```
    book_id = input("Enter Book ID: ")
    title = input("Enter Book Title: ")
    author = input("Enter Author Name: ")
    category = input("Enter Category (Comic/Thriller/Mystery/Fantasy/etc): ")
    row = int(input("Enter Row Number: "))

    books[book_id] = {
        "title": title,
        "author": author,
        "category": category,
        "row": row,
        "available": True
    }

    print("Book added successfully!\n")


# --------------------------------
# 4. Display Library Books
# --------------------------------
def display_books():
    print("\n----- LIBRARY BOOK LIST -----")

    if not books:
        print("No books in library\n")
        return

    for book_id, info in books.items():
        print(f"""
Book ID     : {book_id}
Title       : {info['title']}
Author      : {info['author']}
Category    : {info['category']}
Row         : {info['row']}
Available   : {info['available']}
-----------------------------""")


# --------------------------------
# 5. Search Books by Category
# --------------------------------
```

```
def search_by_category():
    print("\n----- SEARCH BY CATEGORY -----")

    category = input("Enter category: ")

    found = False

    for book_id, info in books.items():
        if info["category"].lower() == category.lower():
            print(info["title"], "-", info["author"])
            found = True

    if not found:
        print("No books found in this category")


# --------------------------------
# 6. Borrow Book
# --------------------------------
def borrow_book():
    print("\n----- BORROW BOOK -----")

    user_id = input("Enter User ID: ")
    book_id = input("Enter Book ID: ")

    if book_id not in books:
        print("Book not found")
        return

    if not books[book_id]["available"]:
        print("Book already borrowed")
        return

    date = datetime.date.today()

    borrow_records[book_id] = {
        "user_id": user_id,
        "borrow_date": date,
        "return_date": None
    }
```

```
    books[book_id]["available"] = False

    print("Book borrowed successfully on", date)


# --------------------------------
# 7. Return Book
# --------------------------------
def return_book():
    print("\n----- RETURN BOOK -----")

    book_id = input("Enter Book ID: ")

    if book_id not in borrow_records:
        print("No record found")
        return

    date = datetime.date.today()

    borrow_records[book_id]["return_date"] = date
    books[book_id]["available"] = True

    print("Book returned on", date)


# --------------------------------
# 8. View Borrow Records
# --------------------------------
def view_records():
    print("\n----- BORROW RECORDS -----")

    if not borrow_records:
        print("No borrow records")
        return

    for book_id, record in borrow_records.items():
        print(f"""
Book ID     : {book_id}
User ID     : {record['user_id']}
Borrow Date  : {record['borrow_date']}
Return Date  : {record['return_date']}
```

```
---------------------------""")


# ==================================
#         MAIN MENU
# ==================================
while True:

    print("""
=====================================
      MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit
""")

    choice = input("Enter your choice: ")

    if choice == "1":
        register_user()

    elif choice == "2":
        setup_library_rows()

    elif choice == "3":
        add_books()

    elif choice == "4":
        display_books()

    elif choice == "5":
        search_by_category()
```

```
    elif choice == "6":
        borrow_book()

    elif choice == "7":
        return_book()

    elif choice == "8":
        view_records()

    elif choice == "9":
        print("Thank you for using Majestic Library System")
        break

    else:
        print("Invalid choice")
```

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8. Sample Output

```
=====================================
      MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 1

----- USER REGISTRATION -----
Enter User ID: U101
Enter User Name: Shalini
Enter Phone Number: 9876543210

User registered successfully!
```

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Enter Book ID: 454545
Book not found

=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 7

----- RETURN BOOK -----
Enter Book ID: 454545
No record found

=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 8

----- BORROW RECORDS -----
No borrow records

=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
```

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
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=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 5

----- SEARCH BY CATEGORY -----
Enter category: fantasy
No books found in this category

=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 6

----- BORROW BOOK -----
Enter User ID: 4116252430
Enter Book ID: 454545
Book not found

=====================================
        MAJESTIC LIBRARY SYSTEM
=====================================
```

```
*IDLE Shell 3.14.2*
File  Edit  Shell  Debug  Options  Window  Help
Enter your choice: 2

----- LIBRARY ROW SETUP -----
Enter number of rows in library: 10
How many books in Row 1: 15
How many books in Row 2: 100
How many books in Row 3: 100
How many books in Row 4: 400
How many books in Row 5: 500
How many books in Row 6: 600
How many books in Row 7: 700
How many books in Row 8: 800
How many books in Row 9: 900
How many books in Row 10: 100
Library rows created successfully!


====================================
        MAJESTIC LIBRARY SYSTEM
====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 3

----- ADD BOOKS -----
Enter Book ID: 45545
Enter Book Title: will she come back
Enter Author Name: shalini
Enter Category (Comic/Thriller/Mystery/Fantasy/etc): romantic
Enter Row Number: 5
Book added successfully!


====================================
        MAJESTIC LIBRARY SYSTEM
====================================
```

```
*IDLE Shell 3.14.2*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ========================================================== RESTART: C:/Users/Shali/Documents/python/mini library system.py ==========================================================

====================================
        MAJESTIC LIBRARY SYSTEM
====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 1

----- USER REGISTRATION -----
Enter User ID: 411625243052
Enter User Name: Shalini
Enter Phone Number: 7339485184
User registered successfully!


====================================
        MAJESTIC LIBRARY SYSTEM
====================================

1. Register User
2. Setup Library Rows
3. Add Books
4. Display Books
5. Search Books by Category
6. Borrow Book
7. Return Book
8. View Borrow Records
9. Exit

Enter your choice: 2

----- LIBRARY ROW SETUP -----
Enter number of rows in library: 10
How many books in Row 1: 15
```

9. Result

The Library Management System successfully performs user registration, book addition, searching, borrowing, returning, and record maintenance operations using Python.

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10. Conclusion

This project demonstrates how Python can be used to develop a simple Library Management System. The program uses dictionaries, loops, and conditional statements to manage books and users efficiently through a menu-driven interface.